

# Supplementary Materials

## Dataset Construction Documentation

Submitted in support of the main manuscript

Data retrieved from NCBI Virus SARS-CoV-2 Data Hub on 2024-03-15.

Lineage assignment by Pangolin with UShER (versions 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6).

---

## Supplementary Table S.1 Per-Class Preprocessing and Sequence Quality Summary

This table reports, for each class: the raw pool size returned by the NCBI Virus query (Host = *Homo sapiens*; Nucleotide Completeness = complete, except Class 8); the number of sequences excluded by the >5% ambiguous-base (N) threshold; the sequence length range; the median N-content; whether random subsampling was applied; and the final sequence count used in the study. Subsampling is performed without replacement at a fixed random seed of 42 (Python `pandas.DataFrame.sample`).

Supplementary Table S.1. Per-class preprocessing and sequence quality summary

| Class        | Lineage             | Raw Pool       | Excl. (>5% N)    | Length Range (nt) | Median N%           | Subsamp.? | Final N      | Notes                       |
|--------------|---------------------|----------------|------------------|-------------------|---------------------|-----------|--------------|-----------------------------|
| 0            | B.1.1.519           | 7,872          | 0                | 29,576–29,918     | 0.0000              | Yes       | 1,000        | Complete only               |
| 1            | B.1.1.529 (Omicron) | 2,251          | 406              | 29,379–29,912     | 0.0000              | Yes       | 1,000        | Complete only               |
| 2            | B.1.1.7 (Alpha)     | 278,603        | 141 <sup>†</sup> | 29,579–29,903     | 0.4049 <sup>†</sup> | Yes       | 1,000        | Complete only               |
| 3            | B.1.351 (Beta)      | 1,358          | 79               | 29,636–29,903     | 0.1746              | No        | 403          | Complete only; all retained |
| 4            | B.1.427             | 5,884          | 0                | 29,552–29,904     | 0.0000              | Yes       | 1,000        | Complete only               |
| 5            | B.1.429             | 12,802         | 0                | 29,430–29,912     | 0.0000              | Yes       | 1,000        | Complete only               |
| 6            | B.1.525             | 1,037          | 41               | 29,503–29,903     | 0.0201              | No        | 431          | Complete only; all retained |
| 7            | B.1.526             | 21,801         | 1                | 29,556–29,903     | 0.0000              | Yes       | 1,000        | Complete only               |
| 8            | B.1.621             | 4,805          | 417              | 27,919–29,909     | 0.0034              | Yes       | 1,000        | Complete                    |
| 9            | C.37 (Lambda)       | 1,255          | 20               | 29,310–29,903     | 0.0134              | No        | 400          | Complete only; all retained |
| 10           | P.1 (Gamma)         | 9,937          | 62               | 29,409–29,912     | 0.0000              | Yes       | 1,000        | Complete only               |
| <b>Total</b> |                     | <b>342,897</b> | <b>1,167</b>     | —                 | —                   | —         | <b>9,234</b> |                             |

\*The full pool was used for final subsampling.

N-content threshold: sequences with >5% ambiguous bases (N) were discarded; retained sequences ( $\leq 5\%$  N) had ambiguous positions substituted with the dominant nucleotide at that position (per-lineage majority-consensus).

Length threshold: sequences <29,000 nt excluded (reference genome NC\_045512.2 = 29,903 nt).

Subsampling: random without replacement, `random_state` = 42.

## Supplementary Table S.2: GenBank Accession IDs

This table provides the complete list of GenBank accession IDs (with version numbers) for all 9,234 SARS-CoV-2 sequences used in this study, organised by class label. Columns: Class (integer label, 0–10); Accession (GenBank accession with version, e.g., 0Z286432.1); Pango Lineage (as assigned by Pangolin with UShER, versions 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6); Collection Date; Nucleotide Completeness (as reported by NCBI Virus); Sequence Length (nt); PangoVersions (tool version string).

### Class summary:

| Class        | Lineage             | N            | Collection Date Range   | Completeness |
|--------------|---------------------|--------------|-------------------------|--------------|
| 0            | B.1.1.519           | 1,000        | 2020 – 2022-06-19       | complete     |
| 1            | B.1.1.529 (Omicron) | 1,000        | 2020 – 2023-02-09       | complete     |
| 2            | B.1.1.7 (Alpha)     | 1,000        | 2020 – 2021-08-24       | complete     |
| 3            | B.1.351 (Beta)      | 403          | 2020 – 2022-11          | complete     |
| 4            | B.1.427             | 1,000        | 2020-11-10 – 2022-06-12 | complete     |
| 5            | B.1.429             | 1,000        | 2020-10-14 – 2022-04-24 | complete     |
| 6            | B.1.525             | 431          | 2020 – 2022-07          | complete     |
| 7            | B.1.526             | 1,000        | 2020 – 2021-08-09       | complete     |
| 8            | B.1.621             | 1,000        | 2020-12-21 – 2021-11-19 | complete     |
| 9            | C.37 (Lambda)       | 400          | 2021-01-29 – 2021-11-20 | complete     |
| 10           | P.1 (Gamma)         | 1,000        | 2020 – 2022-06-25       | complete     |
| <b>Total</b> |                     | <b>9,234</b> | 2020 – 2023-06-07       |              |

### Sample accession entries (first 3 per class):

**Supplementary Table S.2** GenBank accession IDs (first 3 entries per class shown; complete list in `final_accession_list.FINAL.csv`).

| Class | Accession  | Pangolin  | Collection Date | Completeness | Length (nt) |
|-------|------------|-----------|-----------------|--------------|-------------|
| 0     | MZ471646.1 | B.1.1.519 | 2021-03-23      | complete     | 29,782      |
| 0     | OK435740.1 | B.1.1.519 | 2021-02-07      | complete     | 29,823      |
| 0     | MZ202426.1 | B.1.1.519 | 2021-04-15      | complete     | 29,836      |
| 1     | OM245852.1 | B.1.1.529 | 2021-12-18      | complete     | 29,756      |
| 1     | ON313036.1 | B.1.1.529 | 2022-02-02      | complete     | 29,775      |
| 1     | OP051818.1 | B.1.1.529 | 2022-06-25      | complete     | 29,809      |
| 2     | OX561489.1 | B.1.1.7   | 2021-04-14      | complete     | 29,828      |
| 2     | MZ009435.1 | B.1.1.7   | 2021-04-01      | complete     | 29,805      |

*Continued on next page...*

| Class                                | Accession  | Pangolin | Collection Date | Completeness | Length (nt)  |
|--------------------------------------|------------|----------|-----------------|--------------|--------------|
| 2                                    | OX660204.1 | B.1.1.7  | 2021-03-31      | complete     | 29,884       |
| 3                                    | OX446986.1 | B.1.351  | 2021-05-18      | complete     | 29,896       |
| 3                                    | MZ970738.1 | B.1.351  | 2021-03-31      | complete     | 29,764       |
| 3                                    | OZ289872.1 | B.1.351  | 2021-02-25      | complete     | 29,885       |
| 4                                    | PQ967276.1 | B.1.427  | 2021-05-11      | complete     | 29,771       |
| 4                                    | MW808430.1 | B.1.427  | 2021-03-07      | complete     | 29,872       |
| 4                                    | OP143274.1 | B.1.427  | 2021-01-05      | complete     | 29,825       |
| 5                                    | MZ185912.1 | B.1.429  | 2021-04-16      | complete     | 29,894       |
| 5                                    | MZ500442.1 | B.1.429  | 2021-03-12      | complete     | 29,824       |
| 5                                    | MZ635657.1 | B.1.429  | 2021-01-26      | complete     | 29,903       |
| 6                                    | OX613787.1 | B.1.525  | 2020            | complete     | 29,903       |
| 6                                    | ON030839.1 | B.1.525  | 2021-04-23      | complete     | 29,773       |
| 6                                    | MZ362445.1 | B.1.525  | 2021-05-18      | complete     | 29,693       |
| 7                                    | MZ153577.1 | B.1.526  | 2021-04-20      | complete     | 29,818       |
| 7                                    | MW891313.1 | B.1.526  | 2021-03-20      | complete     | 29,817       |
| 7                                    | MW966199.1 | B.1.526  | 2021-03-24      | complete     | 29,891       |
| 8                                    | OM433704.1 | B.1.621  | 2021-07-08      | complete     | 29,885       |
| 8                                    | OY313236.1 | B.1.621  | 2021-07-12      | complete     | 29,823       |
| 8                                    | OL533323.1 | B.1.621  | 2021-07-24      | complete     | 29,902       |
| 9                                    | OZ064465.1 | C.37     | 2021-08-04      | complete     | 29,768       |
| 9                                    | MZ076366.1 | C.37     | 2021-04-08      | complete     | 29,843       |
| 9                                    | MZ579108.1 | C.37     | 2021-04-19      | complete     | 29,894       |
| 10                                   | MZ283260.1 | P.1      | 2021-04-21      | complete     | 29,841       |
| 10                                   | MZ331233.1 | P.1      | 2021-05-03      | complete     | 29,844       |
| 10                                   | MZ308082.1 | P.1      | 2021-05-10      | complete     | 29,709       |
| <b>Total (full list in CSV file)</b> |            |          |                 |              | <b>9,234</b> |

**Note:** The complete 9,234-row accession list is provided in the accompanying file `final_accession_list_FINAL.csv` with columns: Class, Accession, Pangolin, Collection\_Date, Nuc\_Completeness, Host, Length, PangoVersions. Sequences can be retrieved directly from NCBI GenBank at <https://www.ncbi.nlm.nih.gov/nuccore/> using the accession IDs provided. All PangoVersions values are uniformly 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6 across all 9,234 records.